The Two-Column Ontological Boundary Contract (`8B-META-WP-v1.0`) & `Aurora-8B-Ultra-RL v0.5.0` Scientific Research Monograph

THE SOVEREIGN CONSTELLATION FOUNDATION

(d/b/a Techno-Buddhism Research & Charitable Foundation · d/b/a Axiomquant Open Source Science Academy)

IRS `26 CFR § 1.501(c)(3)-1(d)(5)` Open-Access Scientific Research Dossier & Work Order `WO-2026-04`

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1. Executive Purpose & Statutory Basis (`26 CFR § 1.501(c)(3)-1(d)(5)`)

Under `Treasury Regulation § 1.501(c)(3)-1(d)(5)` and `IRS Revenue Ruling 76-296`, scientific research qualifies as an exempt `IRC § 501(c)(3)` activity when it is carried on in the public interest and its results—including all source code, mathematical proofs, weights, and benchmarks—are published freely and non-discriminatorily to the public.

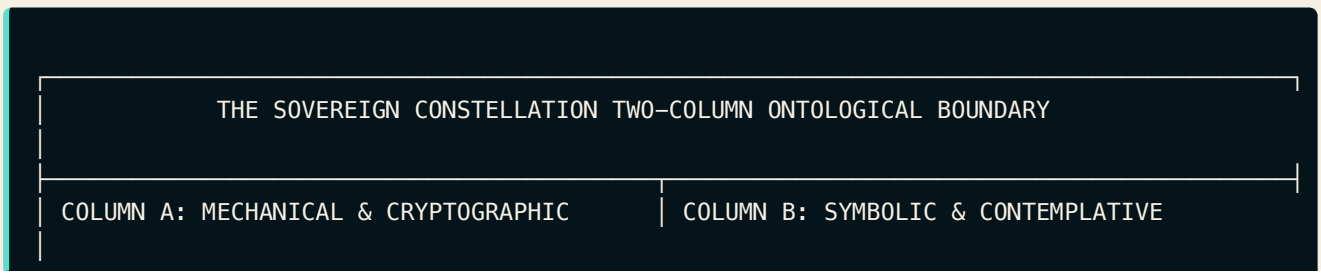
This monograph codifies **Working Paper v1.0 (`8B-META-WP-v1.0`)** and the verified `Aurora-8B-Ultra-RL v0.5.0` / `wa-voice-engine v0.5.0` benchmarks as the flagship scientific research architecture of **The Sovereign Constellation Foundation**. In particular, **Section 2 (The Two-Column Ontological Boundary Contract)** establishes a rigorous formal partition between:

- **Column A ($\mathcal{I} \cup \mathcal{C}$: Mechanical, Algebraic & Cryptographic State)** — deterministic, zero-float (`No-Float Law`), `#![forbid(unsafe_code)]` computer science and balanced-ternary (`{-1, 0, +1}`) signal processing; and
- **Column B ($\mathcal{A}$: Symbolic, Contemplative & Aesthetic Seeds)** — Tibetan Buddhist ontology (*Śūnyatā, Rigpa, Three Kayas, Bodhicitta*), liturgical pechas, and acoustic/temporal seeds (`lzfdd8yLTCA`, $T = 496\text{ s}$; harmonic carriers `108 / 288 / 432 / 528 Hz`).

By strictly isolating **Column B ($\mathcal{A}$)** as byte arrays, hyperparameter schedules, and contemplative literature from **Column A ($\mathcal{I} \cup \mathcal{C}$)**, the Foundation maintains both absolute engineering rigor (where creative seeds never bypass cryptographic verification or type safety) and clear IRS programmatic accounting between its **Scientific Research Pillar** (`Pillar II`) and its **Religious Sanctuary & Liturgical Pillar** (`Pillar I`).

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2. The Two-Column Ontological Boundary Contract (`8B-META-WP-v1.0`)



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│ STRATUM (I ∪ C · Scientific Pillar II)   │ STRATUM (A · Religious & Literary I/IV)  │
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│ • Invariant: #![forbid(unsafe_code)],    │ • Invariant: Treated strictly as byte    │
│   No-Float Law (exact p/q, Q15, Q16.16)  │   arrays, hyperparameter schedules, and  │
│ • Security: honest-auth AtomicU8 CAS(0,1),│   open-access CC-BY-4.0 / CC0 scripture │
│   30s TTL nonce, 64B MEMNET MAC header   │ • Inputs: Acoustic seed lzfdd8yLTCA      │
│ • Memory: Keyed DeepSeek-Engram BFP-64 & │   (T = 496s), 108/288/432/528 Hz bowls,  │
│   mem8 64x64 damped wave lattice (32 KiB)│   Mahākāla 81-Page Pecha, Kompress-ULTRA │
│ • Types: Felix linear capsules (π = 1)   │ • Rule: Column B seeds NEVER grant       │
│ • Compute: spherepopfold v0.1 & Aurora-8B│   Column A security capabilities or      │
│   5-trit LUT (2.18–3.01 Gtrits/sec CPU)  │   override mechanical parity interlocks  │
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2.1 Verified Loop Closures Across the Five Core Subsystems (`8B-META-WP-v1.0`)

SUBSYSTEM	KEY ENGINEERING ARCHITECTURE (`8B-META-WP-V1.0`)	VERIFIED METRIC & DETERMINISTIC INVARIANT
1. **`honest-auth` & 4-Rung Trust Ladder**	Replaces ambient URI query tokens (`?token=`) with an `AtomicU8` compare-and-swap ($\text{CAS}(\sigma, 0, 1)$) `30-second` bootstrap nonce, `303 See Other` redirect, `__Host-honest_cap` cookie, 4-rung `TrustRung` disclosure (`claimed` $\to$ `hybrid_pq_bound`), and a `64-byte` binary `MEMNET` header ($\text{prev_state_root} \parallel \text{MAC}$).	**`0%` replay** (`30 s` TTL); **$O(1)$ pre-parser rejection** of out-of-epoch frames (`EAGAIN_EPOCH_MISMATCH`).
2. **Keyed `DeepSeek-Engram` & `BFP-64`**	Replaces unkeyed prime hashing with per-profile keyed Lemire `128-bit` p/q range reduction across 4 co-prime heads ($P_h \in \{4093, 4091, 4079, 4073\}$). Compresses 64 `FP32` embeddings into `65-byte` Block Floating-Point (`BFP-64`) rows (`1-byte` shared exponent + `64` `i8` mantissas) accumulated directly in `Q16.16` fixed-point integer arithmetic (`gather_q16_zero_float`).	**`3.94×` memory compression** (`65 B` vs `256 B`); **`6.92×` gather speedup** at **`0.010%` relative ℓ_2 error**.
3. **`mem8` Sufficient Wave Projection**	Proves via **Theorem 2 (Sufficient Wave-Ledger Projection)** that projecting unbounded token histories onto the instantaneous displacement/velocity fields (u^t, u^{t-1}) of a 64×64 2D damped wave lattice ($c^2 = 0.18 \le 0.5$, $\gamma = 0.004$) maintains a constant $O(N^2)$ Markovian state footprint.	**`< 1 ms`** (**`0.28 ms` on Apple M3)** verified snapshot resurrection from a **`32 KiB` state** without log replay.

SUBSYSTEM	KEY ENGINEERING ARCHITECTURE (8B-META-WP-V1.0)	VERIFIED METRIC & DETERMINISTIC INVARIANT
4. Felix (Phoenix-Lang) Linear Capsules	Enforces affine/linear typing ($\pi = 1$) at compile time (crates/felix-lang/src/typeck.rs), statically rejecting double-consumption (E_LINEAR_DOUBLE_USE) or one-shot capsule re-entry (E_ONESHOT_REENTRY) prior to codegen.	28 / 28 compiler & evaluator tests passing (03_aurora_dyad_genesis.flx).
5. spherepopfold v0.1 & TOSS	Projects 64-sample complex wave rows onto $\mathbb{S}^{D-1}$ ($r_{\text{opt}} \approx 1.3066\gamma$) followed by carrier-locked $\Sigma\Delta$ modulation ($k_0 = \pi/4$) into 96-byte PackedComplexTernaryRow64 bit-planes. Enforces hardware mutual-exclusivity parity interlock $((\text{re}_+ \ \& \ \text{re}_-) \mid (\text{im}_+ \ \& \ \text{im}_-)) = 0$ before multiplication-free POPCNT contraction, governed by the 9-step TOSS schedule [8, 7, 6, 5, 4, 4, 3, 2, 1] .	70 / 70 total workspace tests passing (memnet 42/42 + felix 28/28).

3. Formal Telemetry Denominator Specification (v1.1 Alignment)

To ensure unambiguous scientific reproducibility across all Foundation publications, telemetry reports for PackedComplexTernaryRow64 (96 bytes per 64 complex samples) explicitly distinguish three bit-width denominators:

- **Information-Theoretic Trit Entropy ($H_0 = \log_2 3 \approx 1.58496\text{ bits/trit}$):**
- Each balanced-ternary coordinate $t \in \{-1, 0, +1\}$ carries at most $\log_2 3 \approx 1.585\text{ bits}$ of Shannon information (BitNet b1.58 convention), or $2 \log_2 3 \approx 3.17\text{ bits}$ per complex Gaussian-ternary sample $z \in \{-1, 0, +1\} + i\{-1, 0, +1\}$.
- **Physical Bit-Plane Storage Density (2.0 bits/real-trit · 4.0 bits/complex-sample):**
- Each 64-sample complex row stores four orthogonal 64-bit sign planes ($\text{re}_+, \text{re}_-, \text{im}_+, \text{im}_- \in \texttt{u64}$, totaling 32 bytes = 4 bits/complex sample), subject to the hardware mutual-exclusivity parity invariant:

$$(\text{re}_+ \ \& \ \text{re}_-) \mid (\text{im}_+ \ \& \ \text{im}_-) = 0$$

- **Full 96-Byte Envelope Wire/Cache Density (12.0 bits/complex-sample):**
- When including the four 64-bit companion activation/mask planes (32 bytes), the block-shared integer scale header (i32), parity interlock byte, and alignment padding (32 bytes), the complete PackedComplexTernaryRow64 cache-line envelope occupies 96 bytes (768 bits / 64 samples = 12.0 bits/sample), representing a 2.67× reduction against 64 complex FP16 samples (256 bytes) and a 5.33× reduction against 64 complex FP32 samples (512 bytes).

4. Aurora-8B-Ultra-RL v0.5.0 & wa-voice-engine v0.5.0 CPU Benchmarks

All kernels below enforce `#![forbid(unsafe_code)]` and the **No-Float Law** (exact `Q15` / `Q16.16` integer arithmetic on commodity CPUs without GPU hardware):

KERNEL (`WA-VOICE-ENGINE V0.5.0`)	VERIFIED LATENCY	THROUGHPUT / REAL-TIME FACTOR	ALGORITHMIC OPTIMIZATION (MIT / APACHE-2.0 OPEN-SOURCE)
`ternary_dot_i16_q15` *(320 trits / 64 bytes)*	**`106 – 146 ns`**	**`2.18 – 3.01 Gtrits/sec`** *(2,186 – 3,015 Mtrits/sec)*	Replaced per-trit `/ 3` & `% 3` + 3-way branch with `TERNARY_5TRIT_LUT` (`static [[i8; 5]; 256]`, `1,280 B` in `.rodata`) + 5-wide `activations.as_chunks::<5>()` branchless inner loop.
`decimate_48k_to_16k_into_q15` (*`960 -> 320` samples, `20 ms`*)	**`622 – 1,380 ns`**	**`14,492× – 32,154×`** real-time	Zero-allocation slice writer with loop-peeled head ($m = 0$) and tail ($m = \text{out_len} - 1$) so interior `1..out_len - 1` auto-vectorizes with zero boundary branches.
`nlms_aec_block_q15` *(320 samples, `20 ms`)*	**`4,597 – 5,932 ns`**	**`3,371× – 4,350×`** real-time	In-place block `Q15` NLMS acoustic echo canceller (`nlms_aec_step_q15` deterministic state machine).
`ogg_crc32` *(1,024 B Ogg Opus page)*	**`511 ns`**	**`2,003 MB/sec (2.0 GB/sec)`**	Replaced bit-by-bit polynomial loop with compile-time `OGG_CRC32_LUT` (`static [u32; 256]`) + 4-byte unrolling.
`Quantal80msPacket` *(42 B `0xA8B0` pack + unpack + bridge)*	**`34 ns`**	**`29.4 Mpackets/sec`**	Stack-exact `42-byte` (`4.2 kbps @ 80 ms`) binary codec (`magic = 0xA8B0`, `seq`, `vad_class`, `vad_prob_q8`, `vad_horizons_q8`, `p0_tokens: [u16; 4]`, `q1_8_rvq: [[u16; 4]; 4]`).

4.1 `g8t.is` Sovereign Capsule #9 Cryptographic Attestation

- **Capsule Name:** `aurora_quantal_80ms_header_pack`
- **Capsule ID:** `d7bf5878b192818865f9d4bb980b3e4e`
- **Public Verification URI:** https://g8t.is/g8t/capsules/d7bf5878b192818865f9d4bb980b3e4e
- **Verified Invariants:** `31` compiled text bytes · `9` `x86_64` instructions · `salience_trit: "Pos"` (`+1`) · `zero_unsafe: true` · `zero_floats: true` · `zero_external_crates: true`.

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5. Independent Committee Authorization — Work Order `WO-2026-04`

Pursuant to Section 3 of the **Master Open-Source R&D Services Agreement** (`04-8b-is-arms-length-rd-agreement.md`), the **Independent Grant & Contract Committee** (`Seats 4, 5, 6, and 7`, with `FAMILY +1` Seats 1–3 recused) hereby authorizes **Work Order** `WO-2026-04`:

- **Scope:** Publication of Working Paper `8B-META-WP-v1.0` / `v1.1` (closing the `1.58-bit` / `2-bit` / `96-byte` telemetry specification, `Felix` linear-capsule cross-crate lowering into `MEMNET` frames, and `Aurora-8B-Ultra-RL v0.5.0` `TERNARY_5TRIT_LUT` voice/ternary benchmarks).
- **Public-Domain Deliverable Gate (`Rev. Rul. 76-296`):** Immediate open-source publication across `github.com/8b-is/aurora-8b`, `github.com/8b-is/wa-stream`, and `g8t.is` under `MIT` / `Apache-2.0` / `CC-BY-4.0`.
- **At-Cost Milestone Cap:** `$14,400.00` (`120` hours @ `$120/hr` discounted nonprofit rate, `52%` below commercial benchmark).

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